

# Capacity vs. Generalization: A Preliminary Study of Vision Transformer Capacity and Cross-Distribution Robustness in Deepfake Detection

**Abstract**— The rapid development of generative models has made synthetic images increasingly realistic, creating challenges for reliable detection systems that must operate beyond the data distribution observed during training. This study investigates whether Vision Transformer (ViT) model capacity is associated with improved robustness under distribution shift in synthetic-image detection. Two substantially different transformer architectures, DeiT-Tiny and ViT-Base, are evaluated under in-distribution conditions. Training and in-distribution testing use StyleGAN-generated and corresponding real face images, while cross-distribution evaluation combines Stable Diffusion v1.5-generated images with real images from the IMDB-Wiki dataset, changing both the synthetic-image generator and the real-image source. Both models achieve near-perfect in-distribution ROC-AUC, with 0.999899 and 0.999950 obtained by DeiT-Tiny across two seeds and 0.998747 obtained by ViT-Base in its single completed seed. However, performance decreases substantially under cross-distribution evaluation, with ROC-AUC values of 0.523710 and 0.511677 for DeiT-Tiny and 0.539869 for ViT-Base. The results indicate a pronounced gap between performance on familiar synthetic-image distributions and robustness to changed generation and real-image sources. Although ViT-Base achieves the highest cross-distribution AUC among the completed runs, the limited number of ViT-Base seeds prevents a strong conclusion regarding the effect of increased capacity. These preliminary findings highlight the importance of evaluating synthetic-image detectors under distribution shift rather than relying solely on in-distribution performance.

**Index Terms**— deepfake detection, synthetic-image detection, Vision Transformer, DeiT, distribution shift, cross-distribution generalization, robustness.

## I. INTRODUCTION

The rapid development of generative artificial intelligence (AI) has enabled the creation of increasingly realistic synthetic images and videos, creating new challenges for the reliable detection of manipulated and AI-generated media [1], [2]. Although modern detectors can achieve high performance on familiar synthetic content, their effectiveness may decline when the data-generating process changes. This makes generalization beyond the generator observed during training an important requirement for practical synthetic-media detection [2], [3].

A major challenge in synthetic-image detection is distribution shift between the generation process used during training and the distribution encountered during testing. Differences in generation mechanisms and the artifacts they produce can cause detectors that perform well on familiar synthetic sources to degrade substantially on unseen ones [3], [4], [5]. Consequently, evaluation under changed generation conditions provides a more demanding assessment of detector robustness than matched-distribution evaluation alone.

The Vision Transformer (ViT) has emerged as a principal architecture in visual recognition as it encodes images as sequences of patches and uses self-attention to learn relationships between image sections [6]. Their applicability in deepfake and synthetic-image detection has broadened, and recent studies have reviewed the use of ViT-based methods in various detection configurations [7]. This presents an opportunity to use ViTs to investigate if detection capacity is related to detection performance and generalization in distribution shifts.

Model capacity may influence generalization beyond the training distribution. Zhang et al. reported that larger ViTs can reduce the performance gap between in-distribution and out-of-distribution data under several types of distribution shift [8]. However, their experiments considered general visual distribution shifts rather than changes in the generation process of synthetic images. Whether increased ViT capacity

provides a comparable benefit in synthetic-image detection therefore remains an empirical question.

The expanding capacity of a model also leads to the increased use of computational power. As a result, the success of detection on its own does not guarantee success in practical implementations, especially in environments where resources are limited. Researchers in deepfake detection have begun investigating the size of models and the amount of computational power needed in addition to detection successes, while lightweight synthetic fake detectors attempt to marry efficiency and detection abilities [9], [10].

Despite growing research on cross-distribution generalization and efficient synthetic-image detection, the relationship between ViT capacity and robustness under changed synthetic-image distributions remains insufficiently characterized. In particular, it is unclear whether substantially increasing ViT capacity produces a practically meaningful improvement under cross-distribution evaluation and whether such an improvement can be observed consistently.

To investigate this question, this research study compares DeiT-Tiny and ViT-Base under controlled experimental conditions. Both models are trained on StyleGAN-based real/fake face data and evaluated on a StyleGAN in-distribution test set and a cross-distribution test set containing Stable Diffusion v1.5-generated images and IMDB-Wiki real images. The cross-distribution evaluation changes both the synthetic-image generator and the real-image source and is therefore treated as a distribution shift rather than a controlled generator-only shift. The study focuses on the relationship between model capacity and the observed change in detection performance between familiar and changed distributions.

The research study addresses three research questions:

- RQ1: How does Vision Transformer capacity affect synthetic-image detection performance under in-distribution evaluation?

- RQ2: How does Vision Transformer capacity affect detection performance under cross-distribution evaluation involving a concurrent change in the synthetic generator and real-image source?
- RQ3: What relationship is observed between increased Vision Transformer capacity and cross-distribution generalization performance?

The main contribution of this preliminary study is an empirical comparison of two substantially different ViT capacity levels under both in-distribution and cross-distribution evaluation. Rather than evaluating synthetic-image detectors only on familiar data, the study emphasizes the performance gap that emerges when the data-generating process changes.

## II. RELATED WORK

### A. Synthetic-Image Detection and Cross-Distribution Generalization

A synthetic-image detection system uses indications such as visual, spatial, and frequency-domain cues to differentiate synthetic images from genuine ones [1], [2]. Nonetheless, the use of images generated by a particular process for training might give rise to features which are pertinent to one generator, thus making it impossible for the detector to be efficient when the generator changes [3]. This point was illustrated by Ojha et al. who proved that fake image detection might fail in instances where the detectors deal with novel image generators [3]. Similarly, Meng et al. studied cross-domain detection and determined that attributes known as artefacts impede generalization in cross-domain detection [4]. Petrželková and Čech also highlighted the detection system's ability to detect images generated in the known process but inability to identify samples from operators that have not been studied previously [5].

Recent studies have extended this problem to diffusion-generated images. FakeInversion examined detection of images from unseen text-to-image models [11], while DRCT investigated generalization across diffusion generators [12]. Revisiting DIRE further considered universal AI-generated-image detection across multiple generative-model families [13]. Other approaches have explored semantic artifacts, guided diffusion-based training, and self-supervised learning as mechanisms for improving generalization [14],[15], [16]. Together, these studies establish cross-distribution generalization as a persistent challenge, but they do not directly characterize how changing detector capacity affects this robustness.

### B. Vision Transformers and Distribution-Shift Generalization

Vision Transformers represent images as sequences of patches and use self-attention to model relationships among image regions [6]. Their application to deepfake detection has expanded, with recent work surveying ViT-based approaches across different detection architectures [7]. However, strong detection performance alone does not establish whether a ViT-based detector generalizes when the synthetic-image distribution changes.

Model capacity is another factor that may influence generalization. Zhang et al. found that larger ViTs can reduce the performance gap between in-distribution and out-of-distribution data under several distribution shifts [8]. Their study, however, considered general visual distribution shifts rather than generator changes in synthetic-image detection. Thus, whether increased ViT capacity provides a comparable robustness benefit under cross-distribution evaluation remains an open empirical question.

### C. Efficient Detection and Research Gap

When implementing practical synthetic media detection, it is essential to take computational cost into account, especially when working with limited resources or in a real-time context. Frick and Petri assessed the detection of deepfakes with respect to computational factors like inference time, memory usage and size of the model [9]. Just like the above-mentioned work, the development of lightweight AI-generated image detection techniques has involved a careful consideration of the relationship between detection capabilities and efficiency [10].

The existing literature therefore provides three relevant observations: cross-generator generalization remains difficult [3], [4], [5], ViT capacity can influence generalization under some distribution shifts [8], and practical detection requires consideration of computational cost [9], [10]. However, the relationship between substantially different ViT capacity levels and robustness under changed synthetic-image distributions has received limited empirical examination. This study addresses this gap by comparing DeiT-Tiny and ViT-Base under matched-distribution and cross-distribution evaluation.

## III. METHODOLOGY

### A. Experimental Research Questions

The experimental design was developed around three research questions concerning detection performance, distribution-shift robustness, and model capacity. RQ1 examines whether the two models differ under the distribution represented during training. RQ2 examines the change in performance when both the synthetic-image generator and real-image source are changed. RQ3 considers whether the substantially larger parameter count of ViT-Base corresponds to an observable improvement in cross-distribution performance.

### B. Dataset Construction

The experiments employ the 140k Real and Fake Faces dataset [17], which contains 70,000 real face images and 70,000 StyleGAN-generated fake face images. The dataset was divided into training, validation, and in-distribution test sets. The training data split includes 100,000 images, among which there are 50,000 real images and 50,000 fake images. The validation split also includes 10,000 real images and 10,000 fake images respectively. For the in-distribution test set there is a separate data sample of 20,000 images.

To evaluate robustness under changed data distributions, a cross-distribution test set was constructed using 30,000 Stable Diffusion v1.5 [20]-generated fake images from the DeepFakeFace dataset [18] and 30,000 real images from the IMDB-WIKI dataset [19]. The resulting test set contains 60,000 balanced images.

Importantly, this cross-distribution evaluation changes two factors simultaneously: the synthetic-image generator changes from StyleGAN to Stable Diffusion v1.5, and the real-image source changes from the StyleGAN dataset's real

images to IMDB-WIKI images. Therefore, the experiment is interpreted as a cross-distribution evaluation involving generator and real-image source shifts, rather than as a controlled generator-only shift.

**TABLE I**  
**DATASET COMPOSITION**

| Split      | Real   | Fake   | Total   | Distribution                 |
|------------|--------|--------|---------|------------------------------|
| Training   | 50,000 | 50,000 | 100,000 | StyleGAN                     |
| Validation | 10,000 | 10,000 | 20,000  | StyleGAN                     |
| ID Test    | 10,000 | 10,000 | 20,000  | StyleGAN                     |
| Cross Test | 30,000 | 30,000 | 60,000  | Stable Diffusion + IMDB-Wiki |

#### IV. MODEL ARCHITECTURES

Two Vision Transformer models with substantially different parameter capacities were selected: DeiT-Tiny [21] and ViT-Base. Both models use a 16×16 patch size and 224×224 input resolution, allowing their detection performance to be compared under a common image representation.

DeiT-Tiny contains approximately 5.52 million trainable parameters, with a hidden dimension of 192 and three attention heads. ViT-Base contains approximately 85.80 million parameters, with a hidden dimension of 768 and twelve attention heads. Both models were initialized using pretrained weights and adapted for binary real-versus-fake classification.

**TABLE II**  
**MODEL CONFIGURATION**

| Model     | Parameters | Hidden Dimension | Attention Heads | Input   |
|-----------|------------|------------------|-----------------|---------|
| DeiT-Tiny | 5,524,802  | 192              | 3               | 224×224 |
| ViT-Base  | 85,800,194 | 768              | 12              | 224×224 |

#### V. PREPROCESSING AND TRAINING

##### A. Image Preprocessing

All images were transformed into RGB format and adjusted to 224×224 pixels. There was no use of central and random cropping anywhere in the process. Model-specific

normalizing was conducted based on individual pretrained model requirements. DeiT-Tiny employed ImageNet normalization, while ViT-Base used its own normalization regime that utilized channel means and standard deviations of 0.5. The same data preprocessing processes were implemented in each of the four experiments, namely training, validation, in-distribution testing and cross-distribution testing.

##### B. Training Configuration

All models were optimized with AdamW (learning rate  $1 \times 10^{-4}$ , weight decay 0.05) using a cross-entropy loss, for a maximum of 10 epochs. The learning rate followed a cosine annealing schedule (CosineAnnealingLR,  $T_{\max} = 10$ ), stepped once per epoch, with no learning-rate warmup. Training used early stopping with a patience of 3 epochs, monitoring validation ROC-AUC; the checkpoint with the highest validation ROC-AUC was retained as the final model for evaluation. An effective batch size of 64 was used. DeiT-Tiny was trained with a batch size of 64, while ViT-Base used a batch size of 16 with gradient accumulation over four iterations. Experiments were conducted using random seeds 42 and 123 for DeiT-Tiny and seed 42 for ViT-Base in the completed runs.

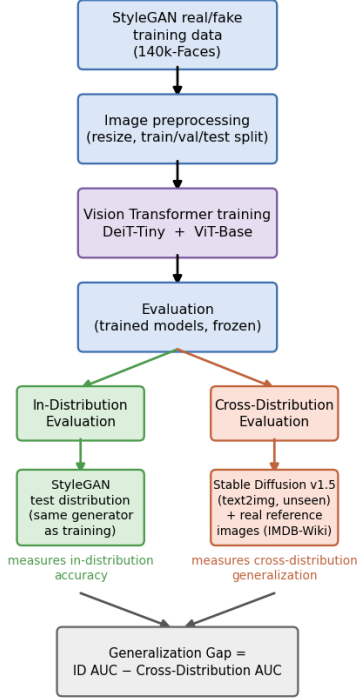
The primary evaluation metric was ROC-AUC because it measures the ability of the detector to rank real and synthetic samples across decision thresholds and is therefore less dependent on a particular classification threshold. Accuracy, precision, recall, and F1-score were also recorded.

**TABLE III**  
**TRAINING CONFIGURATION**

| Setting              | Configuration                                                                        |
|----------------------|--------------------------------------------------------------------------------------|
| Optimizer            | AdamW                                                                                |
| Learning rate        | $1 \times 10^{-4}$                                                                   |
| Weight decay         | 0.05                                                                                 |
| Loss                 | Cross-Entropy                                                                        |
| Scheduler            | Cosine Annealing (CosineAnnealingLR, $T_{\max} = 10$ ), stepped per epoch; no warmup |
| Maximum epochs       | 10                                                                                   |
| Early stopping       | Validation ROC-AUC, patience = 3 epochs                                              |
| Effective batch size | 64                                                                                   |
| Input resolution     | 224×224                                                                              |
| Primary metric       | ROC-AUC                                                                              |

## VI. EVALUATION PROTOCOL

Two evaluation conditions were considered. The first is in-distribution evaluation, in which the model is tested on StyleGAN-based data from the same broad distribution as the training data but using a held-out test split. The second is cross-distribution evaluation, in which the fake images are generated using Stable Diffusion v1.5 and the real images are obtained from the IMDB-WIKI dataset.



The overall experimental pipeline is illustrated in Fig. 1.

Fig. 1. Experimental pipeline for comparing Vision Transformer capacity under in-distribution and cross-distribution evaluation.

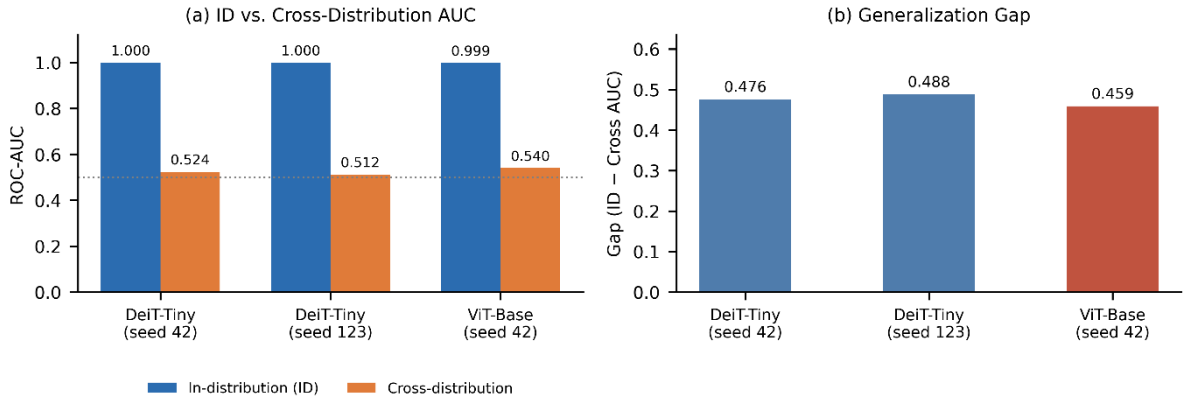


Fig. 2. Comparison of in-distribution and cross-distribution ROC-AUC and generalization gap across the completed experimental runs.

## VIII. ANALYSIS OF CLASSIFICATION BEHAVIOR

The threshold-based metrics provide additional evidence of the difficulty of the cross-distribution condition. For DeiT-

Tiny the difference between the in-distribution and cross-distribution ROC-AUC values is used to quantify the observed generalization degradation. The AUC gap is calculated as the in-distribution AUC minus the cross-distribution AUC. A larger gap indicates a larger decrease in ranking performance under the changed distribution.

## VII. RESULTS

### A. In-Distribution and Cross-Distribution Performance

TABLE IV  
MAIN EXPERIMENTAL RESULTS

| Model     | Seed | ID AUC   | Cross AUC | AUC Gap  |
|-----------|------|----------|-----------|----------|
| DeiT-Tiny | 42   | 0.999899 | 0.523710  | 0.476189 |
| DeiT-Tiny | 123  | 0.999950 | 0.511677  | 0.488273 |
| ViT-Base  | 42   | 0.998747 | 0.539869  | 0.458878 |

The results show that both models achieve near-perfect discrimination under in-distribution evaluation. DeiT-Tiny obtains ROC-AUC values of 0.999899 and 0.999950 across the two completed seeds, while ViT-Base obtains an ROC-AUC of 0.998747 in the completed seed-42 run.

In contrast, both models experience substantial degradation under cross-distribution evaluation. DeiT-Tiny achieves cross-distribution ROC-AUC values of 0.523710 and 0.511677, while ViT-Base achieves 0.539869. These values are close to random-ranking performance, despite the near-perfect in-distribution results.

Among the completed runs, ViT-Base produces the highest cross-distribution ROC-AUC and the smallest observed AUC gap. However, only one ViT-Base seed was completed. Therefore, this result should be interpreted as preliminary evidence rather than as evidence that increased model capacity consistently improves cross-distribution generalization.

Tiny seed 42, the cross-distribution accuracy is approximately 0.501 and recall is 0.991, while precision is approximately 0.500. DeiT-Tiny seed 123 shows a similar pattern, with accuracy of approximately 0.499 and recall of 0.996.

ViT-Base exhibits an even stronger threshold-level bias in the cross-distribution evaluation, achieving accuracy of approximately 0.500, recall of approximately 1.000, and precision of approximately 0.500. Its predictions are overwhelmingly assigned to the fake class.

These observations demonstrate why ROC-AUC is particularly important for interpreting the cross-distribution results. Although threshold-based recall may remain extremely high, the corresponding precision and accuracy indicate that the model does not reliably distinguish the two classes at the default decision threshold. The near-random ROC-AUC values provide stronger evidence of the loss of discriminative ranking ability under the changed distribution.

## IX. MULTI-SEED ANALYSIS

Across the two completed DeiT-Tiny runs, the mean in-distribution ROC-AUC is  $0.999925 \pm 0.000036$ , while the mean cross-distribution ROC-AUC is  $0.517694 \pm 0.008509$ . The mean AUC gap is  $0.482231 \pm 0.008545$ .

The small variation in DeiT-Tiny's in-distribution performance indicates that its near-perfect performance on the familiar distribution is stable across the two tested seeds. However, its cross-distribution performance remains close to random-ranking performance in both runs. This consistency strengthens the observation that the large performance degradation is not specific to a single DeiT-Tiny initialization.

A comparable multi-seed analysis cannot be performed for ViT-Base because only seed 42 was completed. This limitation prevents a reliable estimate of seed-to-seed variability for the larger model.

## X. DISCUSSION

### A. High In-Distribution Performance Does Not Guarantee Robustness

The central observation of the experiment is the large discrepancy between in-distribution and cross-distribution performance. Both models achieve ROC-AUC values close to 1.0 on the familiar StyleGAN distribution, yet both approach random-ranking performance when evaluated on Stable Diffusion-generated images combined with IMDB-Wiki real images.

This finding supports the broader concern identified in previous work that synthetic-image detectors can exploit characteristics associated with the generation process represented in their training data [3], [4],[5]. High performance on a matched distribution therefore does not necessarily indicate that the learned representation captures general characteristics of synthetic imagery.

### B. Effect of Model Capacity

The results provide only preliminary evidence concerning the relationship between ViT capacity and robustness. ViT-Base contains approximately 85.80 million parameters compared with approximately 5.52 million for DeiT-Tiny. In the completed experiments, ViT-Base achieves a cross-distribution ROC-AUC of 0.539869, compared with 0.523710 and 0.511677 for the two DeiT-Tiny runs.

Thus, the larger model achieves the highest cross-distribution AUC among the completed runs. However, the difference is

relatively small compared with the overall degradation from near-perfect in-distribution performance. Furthermore, only one ViT-Base seed was evaluated. The results therefore do not justify the conclusion that increasing ViT capacity consistently improves cross-distribution generalization.

Instead, the findings suggest that increased capacity alone may not be sufficient to produce robust synthetic-image representations under substantial distribution change. Additional experiments involving more model scales, multiple seeds, and controlled generator-only shifts would be required to establish a stronger relationship.

## XI. IMPLICATIONS FOR SYNTHETIC-IMAGE DETECTION

The results have an important implication for the evaluation of synthetic-image detectors. If evaluation is restricted to images generated by processes represented in the training data, detector performance can appear nearly perfect. Such evaluation may therefore provide an incomplete picture of practical robustness.

Cross-distribution evaluation should consequently be included when assessing synthetic-image detection systems. In particular, evaluation should distinguish between performance on familiar generation processes and performance when the generator, image source, or both change.

The observed results also suggest that improving model capacity should not be treated as a standalone solution to distribution-shift problems. Robustness may instead require representations that rely less on source-specific artifacts and more on characteristics that remain informative across different generation processes.

## XII. LIMITATIONS

This research study has several limitations. First, only two ViT capacity levels were investigated, limiting the ability to characterize the relationship between model size and generalization across a continuous range of capacities. Second, multiple random seeds were completed for DeiT-Tiny, but only one ViT-Base seed was completed; consequently, comparisons involving ViT-Base should be considered preliminary.

Third, the cross-distribution evaluation changes both the synthetic-image generator and the real-image source. The experiment therefore does not isolate generator shift from real-image source shift. Fourth, the experiments focus on synthetic face images and may not generalize directly to other image domains or types of synthetic media.

Finally, the study does not establish a causal relationship between model capacity and generalization performance. The observed differences are empirical associations within the tested experimental conditions. Further controlled experiments are required to determine whether increased capacity consistently improves, worsens, or has limited influence on cross-distribution robustness. Potential duplicate or near-duplicate images within and across splits were not explicitly audited.

The contrast between DeiT-Tiny and ViT-Base also includes variations in their initial configuration of regularization and

settings of experimentation such as batch size and gradient accumulation. These processes could lead to differences in performance though the models were performed in the same setting using the same resolution of input data and procedures for optimization.

### XIII. CONCLUSION

The research study focused on examining how capable Vision Transformers are and how synthetic images being detected changes with model performance particularly in in-distribution and cross-distribution scenarios. According to findings, both DeiT-Tiny and ViT-Base had perfect results when detecting in-distribution ROC-AUC, meaning that they were good at detecting StyleGAN-based test distribution. However, both of these models experienced a significant loss of performance when tested on images generated using Stable Diffusion v1.5 in combination with IMDB-Wiki real images.

It can be concluded that there is a considerable difference in terms of the performance results of various models in the case of in-distribution and for cross-distribution performance results. Based on the results of the experiments, ViT-Base has the highest ROC-AUC for all runs done for cross-distribution case but because of the limited number of seeds for ViT-Base testing, it is hard to come to concrete conclusions about the role of the model's capacity in terms of its generalization improvement.

Overall, the findings reinforce the importance of evaluating synthetic-image detectors under distribution change rather than relying solely on matched-distribution performance. Future work should investigate additional model capacity levels, multiple random seeds, controlled generator-only shifts, and approaches designed to learn representations that are less dependent on generator-specific artifacts.

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